

bootstrap [143] underpins our confidence intervals and DeLong [14] gives a closed-form variance for paired ROC-AUC. Our milestone-2 cross-benchmark master table reports per-cell bootstrap intervals; multiple-comparison correction is a follow-up item discussed in §VIII.

K. Adjacent Datasets and Application Domains

Cross-modal generalisation problems often touch zero-shot transfer where classes are unseen at training time [120]; the present work stays in the closed-vocabulary binary anomaly regime, but the reliability-aware fusion idea is in principle portable to that setting. For cyber paired-modality settings the canonical benchmarks are UNSW-NB15 [144], CIC-IDS-2017 [145], and KDD [146], with intrusion-detection methods exploring feature selection [148], [147] and empirical botnet studies [149]. Healthcare AD draws on MIMIC-III [182] and clinically-aware modelling practices [183], [184]. Time-series AD [176], [177], [178] extends the score-level fusion idea to streaming settings. Foundational deep-learning ingredients used throughout this work include ResNet [156], ImageNet pretraining [157], batch normalisation [159], Adam optimisation [158], variational autoencoders [160], and classical VC-dimension theory [162] that justifies generalisation discussions in the thesis appendix. Bahdanau-style soft attention [111] is the precursor of the cross-modal attention layers used here.

L. Dataset Use Matrix

The repository contains many datasets, but they do not carry equal evidential weight. Table II records the current use of each dataset group. This separation is central to the claim boundary: naturally paired, prelocked datasets can carry transfer evidence; opened datasets become development evidence; derived-view and label-aligned datasets remain diagnostic mechanism tests. The full machine-readable audit is maintained in `docs/research/DATASET_USE_MATRIX.md`.

III. SYSTEM AND METHOD

Each sample i contains up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ containing a score, optional confidence, and compact embedding features. A binary mask $m_{i,d}$ indicates whether domain d is missing. The target label is $y_i \in \{0,1\}$ when supervised labels are available.

The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^D), \quad (1)$$

where $\hat{p}_i$ is the final anomaly risk. In static fusion, the learned relationship among domains is fixed after training. In RGA, the system also computes a batch-level reliability vector $r \in [0,1]^D$ at inference time and applies the sample mask to form effective per-sample weights $\tilde{r}_i \in [0,1]^D$. The reliability vector should reduce trust in domains that appear miscalibrated, shifted, uninformative, or missing.

The evaluation problem is not only to maximize ROC-AUC. A useful verification system must also satisfy:

- strong clean ranking and F1 compared with baselines,

- calibration good enough for downstream risk interpretation,
- stability under missing domains,
- graceful degradation under drift and score attacks,
- interpretable domain attribution.

A. ELARA System Architecture

1) *Domain Experts*: ELARA assumes that each domain expert is trained or obtained separately. For the current benchmark, tabular scorers are trained for fraud, cyber, and behavior data, while a text scorer is trained for labeled text. The fusion layer does not require these experts to share the same raw feature space. It only requires score-level outputs and optional compact evidence features.

2) *Fusion Schema*: The unified schema is long-format: each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. This design allows different samples to have different available domains. Missing domains are represented by masks rather than by fabricated raw observations.

3) *Static Attention Path*: The static attention path encodes each available domain into a shared latent space and applies multi-head attention across domains. Domain embeddings allow the model to distinguish evidence sources. Missing domains are ignored through key-padding masks. This is the high-reliability default path.

4) *Reliability-Aware Path*: The reliability-aware path is activated when the average reliability of present domains falls below a configured threshold. This avoids the common failure mode where reliability weights disturb clean predictions even though no domain appears degraded. The reliability path is therefore a stress-response mechanism, not a universal replacement for the static model.

5) *Explanation Path*: For an analyst-facing explanation, the system masks one available domain at a time and recomputes risk. The resulting probability change gives a domain-level counterfactual attribution. This produces statements such as: if text evidence were absent, the risk would decrease by a given amount.

B. RGA Method

1) *Domain Encoding and Attention*: Each domain encoder maps the domain input vector into a shared latent space:

$$h_{i,d} = g_d(x_{i,d}) + e_d + p_d, \quad (2)$$

where e_d is a learned domain embedding and p_d is an optional learned position embedding used by the implementation. A multi-head self-attention block operates across available domain embeddings:

$$z_i = f_\theta(\text{Pool}(\text{Attention}(H_i, m_i))), \quad \hat{p}_i = \sigma(z_i). \quad (3)$$

2) *Reliability Estimation*: After model training, the system fits a reliability estimator on the validation split. For domain d , it stores validation score distributions, fits an isotonic